cs-mail Deployment and Migration Profile

Request Payments and the C-SQD Member Program

C-SQD

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Abstract

An initial C-SQD deployment can operate the communication service and its canonical ledger while one configured payment-processing arrangement executes annual utility charges, request-specific charges, refunds, and member payments. This profile follows a service purchase and a relationship request through to annual member allocation and automatic lump-sum payment to each member's associated bank account. It defines the C-SQD financial program, then explains the operational, privacy, and migration choices needed to deliver it. Company ownership of the pool does not make its member-restricted portion available for ordinary corporate spending.

Design principles. Before changing the implementation or this design, read the Rust-domain design principles (`docs/rust-domain-principles.md`). They are binding for cs-mail and identity-model; new work must not regress them.

1 What This Profile Controls

The *Protocol Specification* defines permission, request outcomes, and the distinction between committed obligations and completed external payments. Section 3, the *C-SQD Financial Program*, adds normative requirements for deployments advertising this profile. In that section, "must" and "must not" are requirements. The remaining sections are deployment guidance. Values explicitly marked proposed are planning defaults, not approved live-payment terms.

The companion C-SQD domain model records the confirmed product rules and deferred decisions. Implementation choices do not override those clarifications. In this profile, a communication *provider* operates the mail service; a payment *processor* executes financial operations. The current product assumes one configured payment-processing arrangement, not routing among multiple processors.

The financial model separates three events: an annual charge purchases service, a conditional charge supports a relationship request, and the member program returns eligible pooled forfeitures through annual allocations. None creates a spendable account balance. The legal characterization and processing arrangement still require assessment; these labels do not establish a money-transmission exemption.

2 A Request from Charge to Outcome

Alice has no standing permission to write to Bob. She selects one of Bob's published request classes and reviews its fixed quote, authorizes $C + S$, and supplies her initial encrypted communication. C-SQD records one request and asks the external provider to capture that charge under a stable

operation key. Once funding satisfies the published finality policy and content is durable, submission places the request before Bob and starts its decision clock.

Bob's follow-up setting governs additional messages during that clock. If he permits them, Alice may add messages under the same request. They cost no more, do not move the deadline, and do not generate another request prompt. If he does not permit them, C-SQD refuses the follow-up without selling another charge against the pending request.

Acceptance records a full refund obligation together with either standing directed permission or a relationship-level express lane. Rejection records processing revenue C and pending forfeiture S. Silence through expiry records C as revenue and a refund obligation for S. Bob receives no personal entitlement from rejecting Alice. Each obligation remains visible until the external execution is reconciled, even if Alice or Bob closes an account.

Funding and submission cannot form one database transaction with a payment provider. Recovery must therefore complete the story. A cancelled or timed-out request being prepared is not submitted when delayed funding evidence arrives; the captured amount is refunded in full. An uncertain capture is queried under its original key rather than retried as another charge. A failed refund remains an obligation and enters retry or review.

3 C-SQD Financial Program

3.1 The annual utility charge purchases service

C-SQD must publish an annual utility price and collect the full charge well before the specified year-long service period begins. An anticipated or finalized pool allocation must not reduce that charge at collection. The later distribution is a separate bank payment; it must not shorten or revoke purchased service.

Published service terms must fix the price, settlement unit, collection date, and year-long service interval. A failed collection and its retry must apply to that same interval and price. A late payment must not shift or extend the period. Policy changes apply to future offers, not an existing purchase. Exact calendar or anniversary anchors and the advance collection lead time remain policy decisions; this profile does not invent a universal date or offset. Collection date, service interval, pool year, and distribution date are distinct facts.

New service coverage requires authenticated evidence that funding has met the provider's finality policy. Pending funding grants no new coverage, and late funding grants no retroactive coverage. Expiry of existing coverage prevents new communication through the local service, including accepted and lane messages; it does not erase relationships, revoke lanes, prevent recipient decisions, or cancel refunds and member payables. A remote sender's service entitlement belongs to that sender's provider.

A billing account and a verified distribution member serve different purposes. An account may purchase service without enrolling in the member program. A member may retain an annual allocation after choosing not to renew service.

3.2 Request pricing is fixed at authorization

The standard request quote combines processing amount $C = \$0.50$ with a recipient-selected collateral amount S for the chosen request class. C-SQD publishes one versioned pricing policy for all of cs-mail: C and the bounds within which every class's S must lie. The bounds are not yet decided; until they are, development uses \$1 to \$25, the range of the former menu. Each recipient address sets S for each of its classes within the bounds; a class outside the current bounds cannot

be quoted until the address republishes. Recipients define the classes and their descriptions; C-SQD does not prescribe purpose-based prices. The sender's selected class and financial terms must remain fixed for the request. A later preference or policy change must not alter that request's signed quote. Both timely acceptance modes must record the full $C + S$ refund obligation with the granted permission; subsequent lane expiry or revocation must not reopen it.

3.3 Ownership and the limits on its use

C-SQD is the service operator and owner of the company pool in this profile. It is responsible for request refunds and finalized member payments, while an external provider executes the underlying transactions. The live policy must identify the contracting entity, merchant/payment responsibilities, and applicable processing arrangement before charges begin.

The pool has no participant ownership shares and no earmarked claim linking a particular rejected request to its recipient. Before annual allocation, a member has no individually finalized claim to a fraction of its unallocated funds. At allocation, the ledger records an individual payment obligation for each qualifying member. This accounting distinction must be reflected in the published program terms rather than inferred from the word "pool."

Eligible forfeitures undergo one disclosed corporate allocation. The remainder is restricted to member distributions. C-SQD must not spend that remainder on operations, payout fees, or corporate losses, or assess it again in later years. Annual utility charges, retained processing revenue, and the corporate allocation support those company costs. The restriction permits no further undisclosed deductions from member funds.

3.4 When a forfeiture becomes eligible

Rejection is the beginning of the forfeiture process, not permission to distribute cash. Each pending lot must retain a scoped request cause, original payment reference, amount, unit, rate/policy version, maturity evidence, and assessment status. Before eligibility, the policy's capture/settlement, refund, dispute, and reserve conditions must be satisfied. Refund obligations and unresolved amounts must remain separately accounted for and unavailable to the pool.

Year end alone does not establish payment finality. A lot that fails maturity at cutoff stays pending until a later eligible year; it does not receive an estimated member allocation. A hold or dispute must identify the affected lot and its review condition. Reserve methodology, release conditions, and later-loss funding must be approved before live distribution, not invented by a worker.

C-SQD must record later external reversals through authorized compensating entries and fund the resulting obligations without silently taking restricted member funds or cancelling finalized payables. A final request outcome remains part of the journal even when a financial correction is necessary.

3.5 The corporate allocation happens once

The current corporate-rate default is $\alpha = 0.03$. The adopted rate and rounding policy must be disclosed through immutable request terms. A future rate change applies prospectively; the rate at a later maturity date does not replace the one attached to an existing request.

For allocation year y, group newly eligible, previously unassessed lots by settlement unit and quoted rate version v. Let F_{yv} be a group's amount in integer minor units. The corporate amount

and new member contribution are

$$G_y = \sum_v \lfloor \alpha_v F_{yv} \rfloor, \qquad M_y = \sum_v F_{yv} - G_y.$$

The floor is applied once per rate cohort and unit at the finalized annual boundary, with any fractional corporate unit left for members. Each contributing lot must be marked assessed in the same atomic allocation record. Replays cannot assess it again. Calculations must use checked integer or exact rational arithmetic, never floating-point currency.

For example, 10,000 units of newly eligible forfeitures at 3% produce 300 of corporate revenue and 9,700 for members. This assessment excludes C, refund obligations and unresolved request amounts. Previously assessed member funds and finalized unpaid allocations are also excluded. The sender is not charged an additional 3%.

3.6 Who qualifies for an equal share

The program benefits qualifying active members generally. It must not weight a share by requests received, collateral contributed, rejections, messages sent, or correspondence with a particular sender. One qualifying member receives one share, irrespective of aliases, mailboxes, or devices.

The distribution identity is distinct from the protocol's private principal and public communication identities. The program must prevent duplicate enrollment while keeping identity evidence out of delivery, relationship, and telemetry records. Eligibility requires program opt-in, the published membership/activity conditions, and required identity checks. Confirmed abuse or duplicate membership must be handled under a disclosed review and correction procedure.

The proposed initial eligibility policy is membership for at least 30 days by cutoff and intentional authenticated use on at least two distinct days of the year. Reading can count. Background synchronization, automated polling, and incoming lane traffic alone do not establish activity. Store minimal activity evidence, not message content or a history of correspondents. The current C-SQD account model is personal: one identifiable person, one account, potentially multiple email addresses, with identity tied to a verified bank account. Accounts whose enrollment review approved it may share a bank account; each keeps its own funding source. Organizational account models are outside this product scope. Renewal for the following service year must not be an eligibility condition. Eligibility is determined from the completed allocation year, independently of the member's billing account.

3.7 From the annual cutoff to fixed payables

Allocations occur once per calendar year. The program must publish its eligibility policy version and cutoff rule before the period. Year y is the half-open interval from January 1 of y to January 1 of $y+1$, in UTC; an event at the ending boundary belongs to the new year. A review interval after cutoff can resolve already recorded evidence without admitting later activity into the closed year.

At finalization, fix the qualifying membership snapshot and eligible funding manifest. Let K_{y-1} be unallocated, previously assessed member carryforward, and let N_y be the qualifying member count. Together with the newly assessed member contribution M_y, the amount available is $D_y = K_{y-1} + M_y$. For $N_y > 0$,

$$a_y = \lfloor D_y / N_y \rfloor, \qquad K_y = D_y - N_y a_y.$$

Each member receives a payable of a_y minor units. The remainder K_y remains restricted carryforward. If $N_y = 0$, no individual allocation is created and $K_y = D_y$. An empty pool creates no payment. Settlement units must be handled separately; no implicit foreign exchange is permitted.

The snapshot, manifest, assessment marks, payables, remainder, and journal entries must finalize as one visible outcome. An operator must not silently rerun the year with a different membership set. Corrections require a separate authorized record with a stated cause and funding source; they do not overwrite the finalized allocation.

3.8 The member's annual distribution date

Finalization establishes each member's share of the previous calendar year's company-wide collateral pool. One ordinary annual distribution covers that share. Published distribution terms set its due date separately from service collection and coverage. Ordinarily distribution occurs during a service year whose utility charge was already paid. Renewal for the next year, and a pending or failed collection for that year, must not delay, reclassify, or cancel the distribution. This profile does not require payment a fixed number of days after collection.

A due date is an earliest execution date, not a guarantee of processor completion that day. After finalization and when due, C-SQD automatically forwards the distribution to the verified bank account associated with the member's account. No separate per-distribution instructions or independently chosen destination are required. Closure must preserve a minimal route for outstanding distributions and refunds. Detailed account management and bank replacement remain deferred.

3.9 One payment, with rebate and excess reporting

Let D_i be member i's finalized allocation and U the annual utility charge ($120 in the clarified example). The reporting classifications are

$$R_i = \min(D_i, U), \qquad X_i = \max(0, D_i - U), \qquad R_i + X_i = D_i.$$

R_i is the utility-fee rebate/refund classification and X_i is the excess cash classification. They are bookkeeping and reporting distinctions within *one lump-sum bank payment*. C-SQD must not implement them as an original-charge refund and a separate payout. Remaining processor-refundable capacity does not enter the calculation, and future renewal status does not change U.

For $U = \$120$, allocations of $80, $120, and $170 produce single payments of those amounts, with rebate/excess breakdowns of $80/$0, $120/$0, and $120/$50 respectively. A zero allocation needs no external payment. Actual relationship-request refunds retain their capture-linked refund meaning.

The distribution has one obligation and one payment lifecycle, including its sequential attempts and authenticated evidence. An uncertain response must be reconciled under the original operation identity. A definite failure may permit a new attempt for the same obligation. A later reversal preserves evidence and restores the unpaid obligation for reconciliation; it must not silently trigger an unrelated payment or erase the allocation.

Failed or delayed payments stay assigned to their members, outside the next year's equal split. Account closure and year rollover do not erase them. C-SQD must not deduct payout fees from the stated allocation. No member-selected payout minimum, cross-year aggregation, or payment-threshold rule is adopted here.

Statements show the full annual utility charge, finalized allocation, rebate and excess classifications, and one distribution completion status. There is no spendable cs-mail balance, transfer to another member, or automatic netting against the next service purchase. The distribution leaves funded coverage intact.

3.10 Published policy and program closure

The launch policy must publish the adopted corporate rate, eligibility thresholds, maturity and reserve criteria, service periods and advance collection terms, distribution terms, and review timelines. It must also define account closure, unclaimed obligations, residual restricted funds, and orderly program termination before collecting real value. Stopping the program cannot by itself convert member-restricted funds into unrestricted corporate revenue or extinguish unpaid obligations.

The program should publish aggregate reconciliation: new eligible forfeitures, corporate allocation, member contribution, carryforward, qualifying member count, finalized payables, and payment completion. Individual correspondence or the amount a particular recipient rejected must not be disclosed through those reports.

4 Operating the Service Without a Wallet

C-SQD can initially run one provider and ledger, but identity, relationship, ciphertext, accounting, membership verification, and telemetry need distinct credentials, keys, and access roles. The payment adapter receives scoped payment references and required amounts. It does not need message content, a recipient's contact graph, or events for every follow-up message.

The operational ledger should reconcile utility-service receipts, request funding, unresolved requests, refunds, pending forfeitures, corporate revenue, restricted funds, and member payables against external statements. An operator investigates mismatches through narrowly authorized records and compensating commands, not arbitrary balance edits. Unknown outcomes stay visible until resolved. A processor outage must not become a duplicate charge or a claim that a refund has completed.

Funding attempts require C-SQD-controlled capacity shared by the account's request and utility charges. The current limit bounds simultaneously unresolved attempts, not lifetime attempts. Failure or reversal restricts the source until authenticated re-verification; bank replacement policy remains deferred. A queued reservation does not authorize dispatch: the durable transaction must check current restrictions before granting a new dispatch. Looking up an uncertain operation under its original identity is reconciliation; dispatching another pull is a new attempt and requires available authority and capacity. Users must not bypass restrictions by creating new requests or changing command identifiers.

Current C-SQD enrollment binds one identifiable person to one account and its verified bank association, with multiple email aliases permitted. Account actions require explicit grants checked against active operational keys; communication permission alone is not financial authority. This does not introduce a separate user-managed billing credential. Bank, membership, and communication records retain separate purposes, access, and scoped identifiers. Legacy organizations may send through express lanes without becoming C-SQD account holders. Future provider migration must preserve recipient-scoped abuse continuity without exporting a universal principal identifier.

5 The Recipient and Sender Experience

The sender sees the recipient's request classes and their collateral requirements, then the chosen class, price, and initial communication for one request, the possible refunds, and the decision window. The interface distinguishes preparation for submission, waiting for the recipient's decision, and the terminal outcome, with external refund status shown separately. It does not present available or

withdrawable funds. A request still being prepared offers authenticated cancellation; closing the application does not imply that the charge was cancelled.

The account view shows the annual service price, coverage interval, next charge date, and scheduled distribution date. It presents the full charge and later lump-sum distribution as separate transactions, with the rebate/excess breakdown as reporting. A member who does not renew can still inspect and receive a finalized allocation.

The recipient sees the selected class, one grouped request, and choices to accept with standing permission, accept with an express lane, reject, or block. The explanation starts with future permission, then the financial consequence. Rejecting does not earn the recipient that request's collateral. Dismissing the prompt is not a decision. Existing late standing acceptance does not reopen the closed request's refund outcome. Bounded-acceptance edge cases remain open in the domain model.

Follow-up controls belong beside request preferences and are enforced at admission. Recipients may disable follow-ups or set bounded counts and rates; changes affect subsequent admissions, not immutable financial terms. Accepted communication and valid express lanes retain their own authorization paths.

The proposed request policy uses a 30-day decision window, a 30-day cooldown after expiry, and a 90-day cooldown after rejection. Cooldowns run from the canonical decision deadline for expiry and the canonical timely rejection/block event for rejection allocation, subject to any later principal-history bound. These values remain proposals. Switching classes cannot bypass a pending request or principal-recipient history. A finalized policy must define recipient-imposed longer limits, explicit invitations that can shorten ordinary cooldowns, and behavior after revocation. Blocking always requires an authenticated unblock; an invitation cannot silently remove it.

6 SMTP and Express-Lane Migration

SMTP is a visibly identified compatibility route because its gateway may handle plaintext. A native message's end-to-end confidentiality must not be claimed for ordinary gateway-readable mail.

Within the funded service boundary, inbound processing checks blocking, standing acceptance, and an applicable recipient-granted lane. It then checks whether an authenticated message may join an identified request awaiting a decision. Only an eligible new request reaches the payment invitation flow. The web authorization binds the directed parties, request, selected request class, initial content/declarations, amount, immutable terms, and deadlines. It never asks the sender to top up a general account.

Subsequent legacy mail must prove its association and satisfy current follow-up policy; an address or subject-line match alone is insufficient authority. A failed lane or follow-up cannot silently authorize a new charge. Invitations need rate limits, anti-enumeration controls, clear expiry, and abuse reporting. Outbound downgrade must be disclosed before the sender commits to that route.

Express-lane messages carry no request charge or pool contribution. Acceptance with a lane settles its request; later lane changes do not reopen that outcome. A lane concerns the relationship and is usually not specific to one conversation. A separate lane-management grant also resolves a pending request with the full refund; terminal settlements remain fixed. Domain-authentication evidence supports the lane's sender scope, not the truth of its declared purpose or a member's active-use eligibility.

7 Incentives, Review, and Later Federation

Equal member distributions remove a direct rejection bounty, but they do not remove economic incentives. Misclassification faces ordinary rejection, without a separate penalty or reputation mechanism. C-SQD earns C on unsuccessful requests and a share of eligible forfeitures. It should publish pricing controls and preserve signed decision evidence so neither the service nor a recipient can manufacture expiry by hiding acceptance. Review should address duplicate members, fabricated activity, abusive request patterns, collusion, and manipulated cutoff evidence.

A single company-owned pool is the initial model. Independent-provider federation requires a new agreement assigning the merchant, refund debtor, pool owner, membership authority, duplicate-member controls, clearing, default losses, and payment responsibility. Inter-provider clearing may net mutual obligations; it cannot create new ownership rules or turn a rejected sender's collateral into a direct recipient payout. The centralized draft therefore does not promise that the former recipient-collateral clearing example transfers unchanged to federation.

8 A Staged Launch

Extend the current deterministic model with request classes and acceptance with a lane, preserving the unresolved interactions listed in the domain model. Validate durable ordering, double-entry obligations, and recovery through a simulated external provider. A desktop pilot can use nonredeemable amounts while testing real encryption, request/follow-up permission, deadlines, and failure behavior.

Before live value, the service needs approved payment and program policies, external reconciliation, capture-after-cancellation recovery, refund retries, maturity controls, reproducible annual snapshots, and failed-payout handling. The first distribution should be demonstrable end to end with advance collection, fixed-period retries, a nonrenewing or closed account, duplicate processor events, and lost-response recovery for one lump-sum payment. Rebate/excess examples must not produce separate operations or completion states. Privacy tests must cover verification, activity evidence, payment records, and backups as well as messages.

SMTP and independent-provider federation are later operational gates. Each gate needs measurable exit evidence appropriate to its risks; a working communications pilot is not evidence that live financial operations or cross-provider pool governance are complete.